



Python Programming Foundations

Week 5 - Files, Errors, & Data Persistence
Day 3: "Comparing Data Representations & Application Structure"

1

Same Information,
Different Forms

CSV shape helps with rows and columns.

JSON shape helps with labels and nested information.

Different shapes make different tasks easier.

2



One Study Task, Four Ways

Same information, different representations

Plain note

Task: Read chapter
Topic: Files
Minutes: 30

CSV row

Task	Topic	Minutes
Read chapter	Files	30

JSON object

```
{  
  "task": "Read chapter",  
  "topic": "Files",  
  "minutes": 30  
}
```

Table row

Task	Topic	Minutes
Read chapter	Files	30

 Different formats, same meaning.
Knowing how data can be represented helps you read it, work with it, and explain it.

Same Information, Different Forms

3

Review: Structure Data Has Shape

- ▶ CSV shape helps with rows and columns.
- ▶ JSON shape helps with labels and nested information.
- ▶ Different shapes make different tasks easier.



4

Structured Data Has Shape

CSV
rows and columns

Task	Topic	Minutes
Read chapter	Files	30

 good for similar records

JSON
labeled grouped values

```
{
  "Task": "Read chapter",
  "Topic": "Files",
  "Minutes": 30
}
```

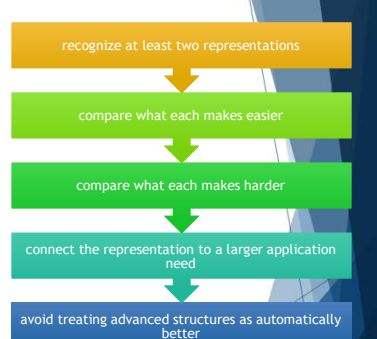

 good for labeled structure

💡 Different shapes make different tasks easier.

Shape of Data (Structured)

5

What Counts as Success Today



```

graph TD
    A[recognize at least two representations] --> B[compare what each makes easier]
    B --> C[compare what each makes harder]
    C --> D[connect the representation to a larger application need]
    D --> E[avoid treating advanced structures as automatically better]
  
```

6

Data Representation Success Pattern

Compare forms using the same information.

1. recognize forms

How can this information be represented?

CSV • JSON • Table • Note • Others

2. what gets easier?

What tasks become simpler with this representation?

- Read
- Filter
- Summarize
- Update

3. what gets harder?

What tasks become more difficult with this representation?

- Look up by key
- Add new fields
- Handle variation

4. larger apps need

What will the larger application need to do?

- Speed
- Flexibility
- Integration
- Maintenance

Task

Read chapter

Topic

Files

Minutes

30

Representation choice depends on use.

Today's Success

7

```
mirror_mod = modifier_obj
mirror_object = mirror
mirror_mod.mirror_object =
operation == "MIRROR_X"
mirror_mod.use_x = True
mirror_mod.use_y = False
mirror_mod.use_z = False
operation == "MIRROR_Y"
mirror_mod.use_x = False
mirror_mod.use_y = True
mirror_mod.use_z = False
operation == "MIRROR_Z"
mirror_mod.use_x = False
mirror_mod.use_y = False
mirror_mod.use_z = True
selection at the end - add
obj.select-1
for obj in scene.objects:
    context.scene.objects.active = obj
    ("Selected" + str(modifier
mirror_object.select = 0
bpy.context.selected_obj
data.objects[one.name].sel
int("please select exactly
--- OPERATOR CLASSES ---
types.Operator):
    x mirror to the selected
object.mirror_mirror_x"
error X"
context.active_object is no
```

What We Will Use Today

- ▶ plain text
- ▶ CSV
- ▶ JSON
- ▶ list/dictionary structures
- ▶ table-like or model-like previews

We will ask what each form makes easier or harder.

8

TODAY'S COMPARISON SET

Some information shown in five sample forms.

plain text

Task: Read chapter
Topic: Files
Minutes: 30

CSV

Task: Read chapter
Topic: Files
Minutes: 30

JSON

Task: Read chapter
Topic: Files
Minutes: 30

list/dictionary

Task: Read chapter
Topic: Files
Minutes: 30

table-like preview

Task: Read chapter
Topic: Files
Minutes: 30

What becomes easier?

Which tasks are simpler or more natural in some forms?

What becomes harder?

Which tasks are trickier or require more work in some forms?

REPRESENTATION CHOICE DEPENDS ON USE.

Toolbox

9

3

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What We Will “Skip” For Now

- ▶ Building a database
- ▶ Writing SQL
- ▶ Implementing an ORM
- ▶ Designing a production data model

Today is recognition and comparison.



10

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Scope Boundary

Focus on what helps us most right now.

1 Lane

Today: recognize and compare

plain text

CSV

JSON

Instructionary

table-like preview

Goal: Understand the shapes and how they compare.

2 Lane

Later: build databases or ORMs

Design schemas, relationships, and ORMs when we start building larger applications.

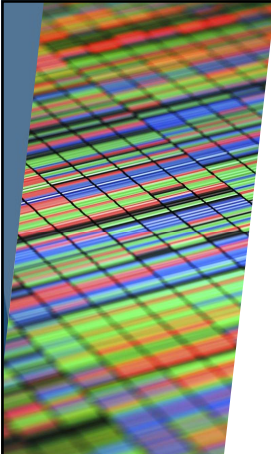
Saved for later: Important and useful when we need it.

We focus on the next right step without losing sight of what comes next.



Parked For Later

11



Data Forms

12

Plain Text Is Human Friendly



Plain text is easy for humans to read.



It becomes harder when the program needs to reliably select fields, filter records, or compare values.

13

Plain Text



Easy for humans

Task: Read chapter
Topic: Files

vs.



Harder for programs
to select fields

Read chapter
Files



Plain text is useful, but structure is limited.

“Human” Friendly:
Plain Text

14

CSV Is Table Friendly

CSV works well when data fits rows and columns.

It is useful for lists of similar records, but it can get awkward when data needs deeper nested structure.

15

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CSV

Comma-Separated Values

Task	Topic	Minutes
Read chapter	Files	30
Review notes	Data shapes	20
Practice examples	Algorithms	45

Good for similar records
Works well when each row has the same fields.

Awkward for deeply nested data
Hard to represent complex or hierarchical relationships.

“Table” Friendly: CSV

16

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JSON Is Structure Friendly

JSON works well when data needs labels, groups, or nested pieces.

It can be very useful, but it can also become hard to read if the nesting gets too deep.

17

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JSON

JavaScript Object Notation

```
{
  "task": "Read chapter",
  "topic": "Files",
  "minutes": 30,
  "status": "planned",
  "details": { ... }
}
```

Good for labeled groups
Names make the data easy to understand and select.

Can get hard to read if nesting grows
More levels add detail but can reduce clarity.

“Structure” Friendly: JSON

18



Code Structures Can Mirror Data

Lists, dictionaries, and simple classes can mirror the shape of stored data.

The program's structure often follows the data's structure.

19

Code Can Mirror Data Shape

Stored Data

task record

task:	Read chapter
topic:	Files
minutes:	30

1. dictionary

```
{
  "task": "Read chapter",
  "topic": "Files",
  "minutes": 30
}
```

2. list of dictionaries

```
[
  { "task": "Read chapter", "topic": "Files", "minutes": 30 },
  { "task": "Review notes", "topic": "Data shapes", "minutes": 30 },
  { "task": "Practice examples", "topic": "Algorithms", "minutes": 45 }
]
```

3. simple class

```
class TaskRecord:
    def __init__(self, task, topic, minutes):
        self.task = task
        self.topic = topic
        self.minutes = minutes
```

The program's structure often follows the data's structure.

Code Structures Can Mirror Data

20

Better Depends On Use

? Ask:

What needs to be read?

What needs to be searched?

What needs to be updated?

What needs to be explained?

The answers guide the representation choice.

21



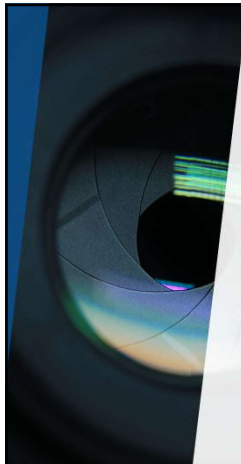
Data Representation Decision

Answer these questions to guide your choice.

- 1 What needs to be read?
- 2 What needs to be searched?
- 3 What needs to be updated?
- 4 What needs to be explained?

Better Depends On Use

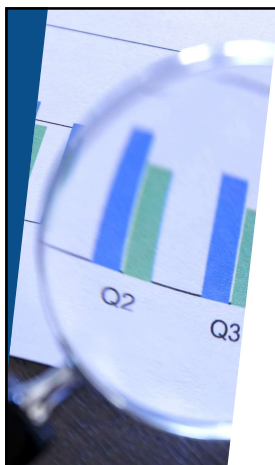
22



Demo

"Basic Type Awareness"

23

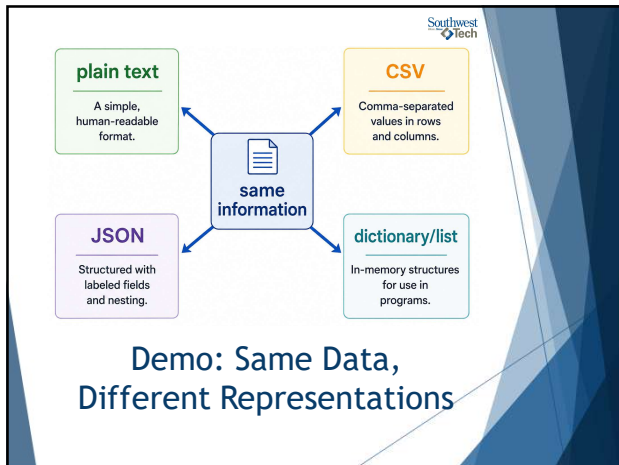


What to Watch For

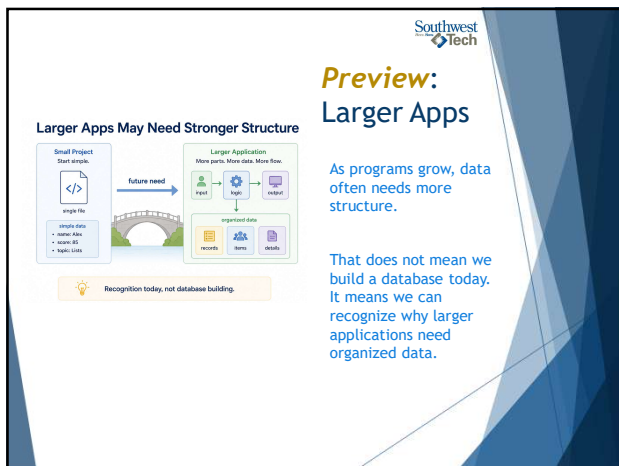
Watch what changes when the same information appears as:

- ▶ plain text
- ▶ CSV-style data
- ▶ JSON-style data
- ▶ dictionary/list structure

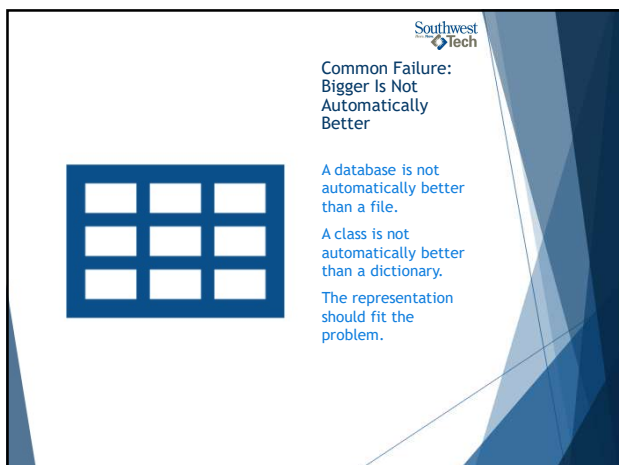
24



25



26



27

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Right-Sized Beats Advanced-Looking

Advanced-Looking Option
too much for this problem

Powerful tools, but more than we need right now.


database
Great for complex relationships and large data.


class
Useful for complex behavior and many features.


API
Helps connect services and external systems.

Great choices for bigger, more complex apps.

vs.

Right-Sized Option
fits the task

Simple, clear, and easy to work with.


single file
Store data in a simple text, CSV, or JSON file.


dictionary
Use a dictionary or list in memory for the data.

Right-sized keeps code and data easy to manage.

 **Choose the representation that fits the problem.**
Simple now. More structure later if (and when) needed.

Common Failure: Bigger Is Not Automatically Better

28

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Assignment #10

For Assignment 10:

- ❖ Compare at least two representations of the same information.

Explain:

- ☐ What each form makes easier,
- ☐ What it makes harder,
- ☐ Where a larger application might need more structure.

29

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A10: Compare and Explain

Compare two representations of the same information and reflect on the differences.

1 

Representation 1

2 

Representation 2

3 

What became easier?

4 

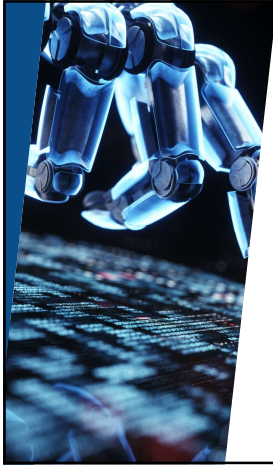
What became harder?

5 

Where might a larger app need more structure?

Assignment 10

30

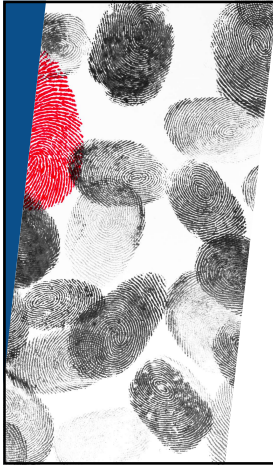


AI Use In Lab 0n

a



31



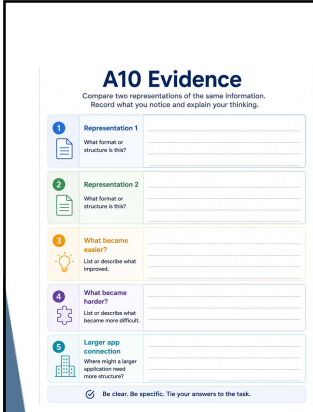
Evidence For A10



Your response should include:

- ▶ at least two representations
- ▶ clear easier/harder comparison
- ▶ beginner-level vocabulary
- ▶ connection to a larger application need
- ▶ AI-use note if AI helped explain a term

32



A10 Evidence

Compare two representations of the same information. Record what you notice and explain your thinking.

1 Representation 1

What format or structure is this?

2 Representation 2

What format or structure is this?

3 What became easier?

List or describe what improved.

4 What became harder?

List or describe what became more difficult.

5 Larger app connection

Where might a larger application need more structure?

☒ Be clear. Be specific. Tie your answers to the task.



Evidence

33



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Closing Success Check

By the end of today, you should be able to say:

"This representation is useful for this purpose, but another representation may be better for a different purpose."

34



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Thank you! Have Fun!

35
